

Complete Radiomics Research Pipeline

Advanced Analysis Modules (2020-2024)

1. Radiogenomics Analysis

2. Deep Radiomics & Feature Learning

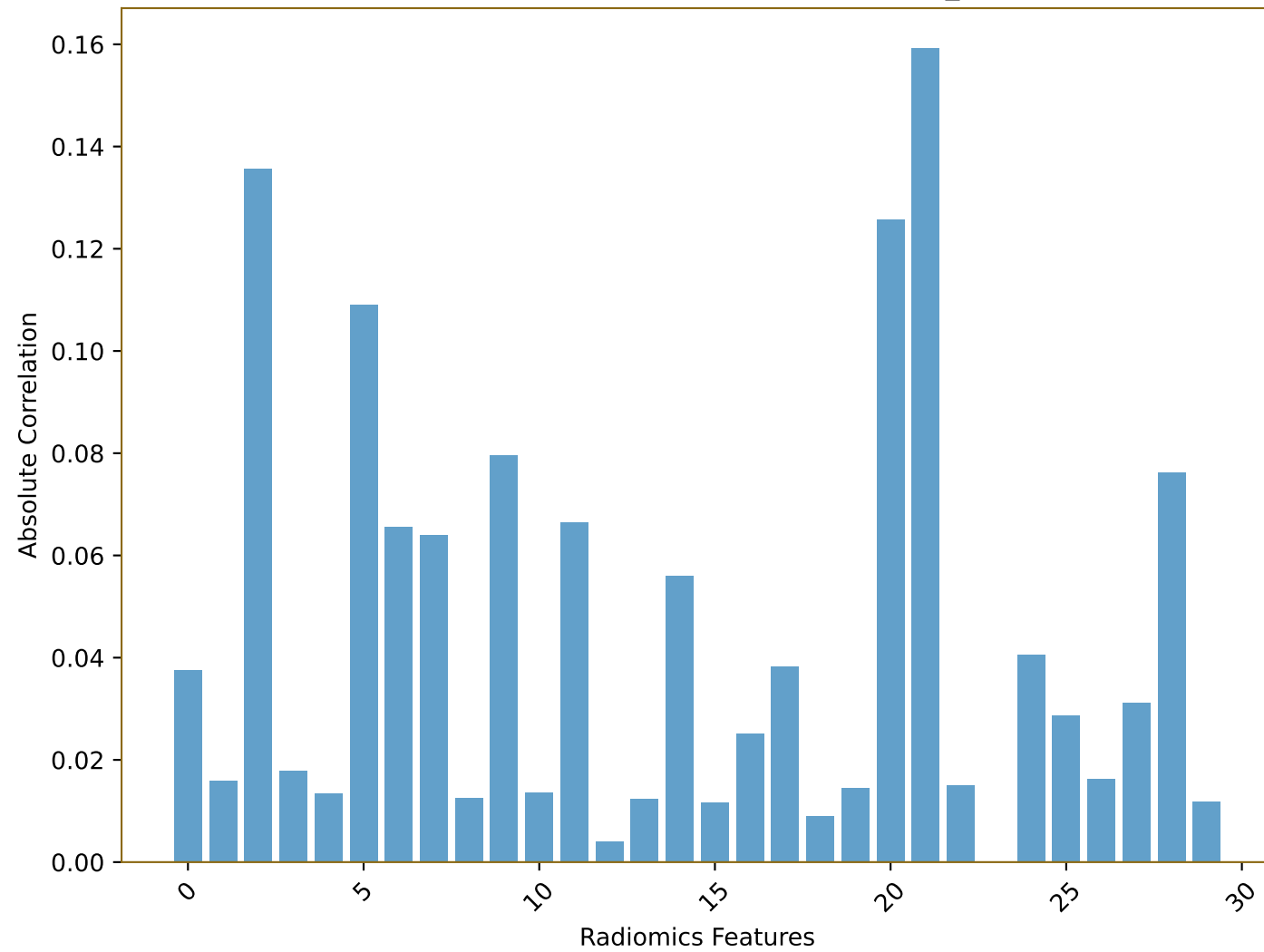
3. Survival & Time-to-Event Analysis

- 4. Radiomics Signature Development
- 5. Explainable AI & Model Interpretability
- 6. Multi-Modal Data Integration
- 7. Clustering & Phenotyping
- 8. Longitudinal & Delta Radiomics
- 9. Feature Robustness & Harmonization
- 10. Clinical Trial Integration

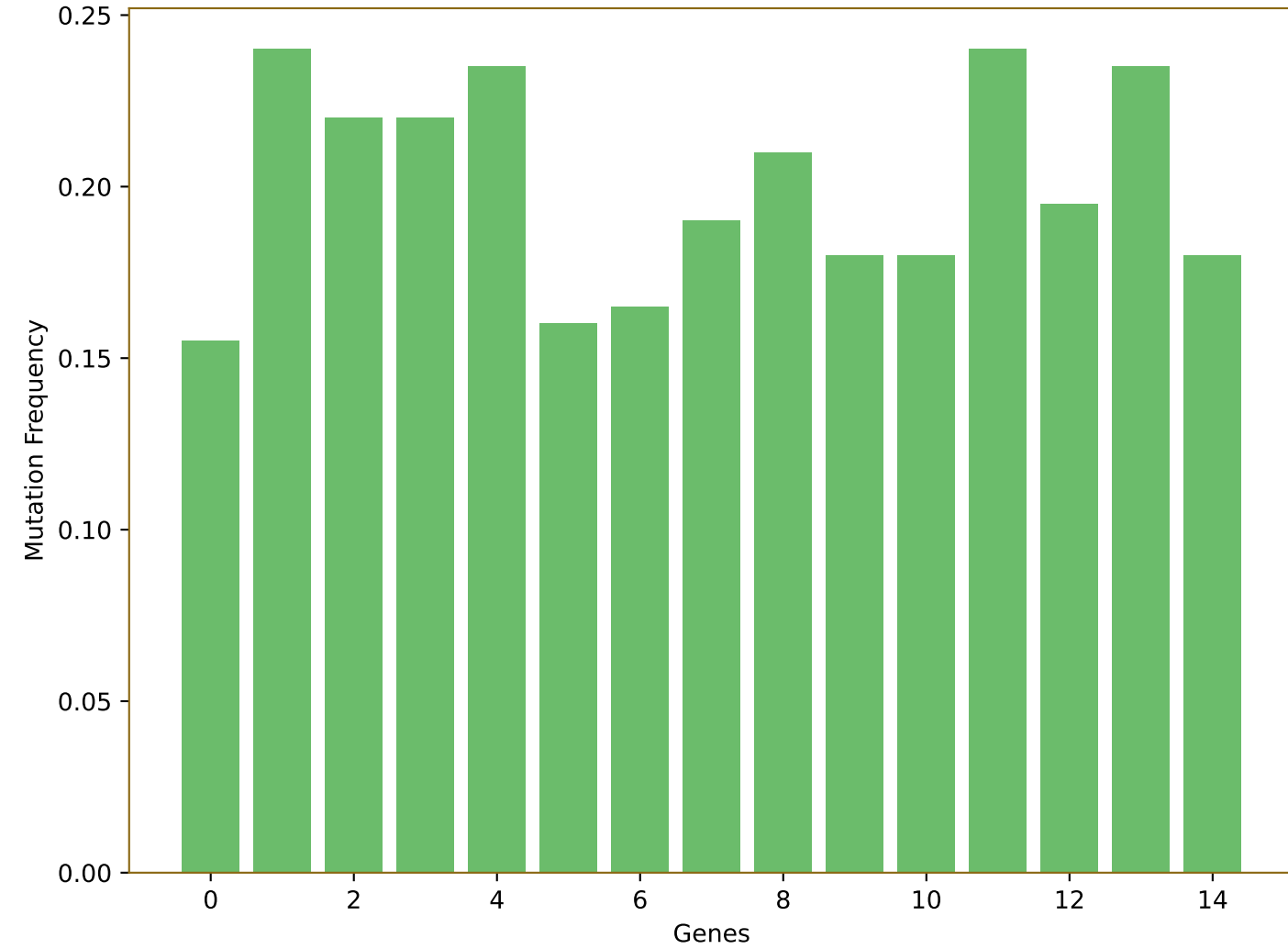
Dataset: 200 patients, 31 radiomics features

Radiogenomics Analysis

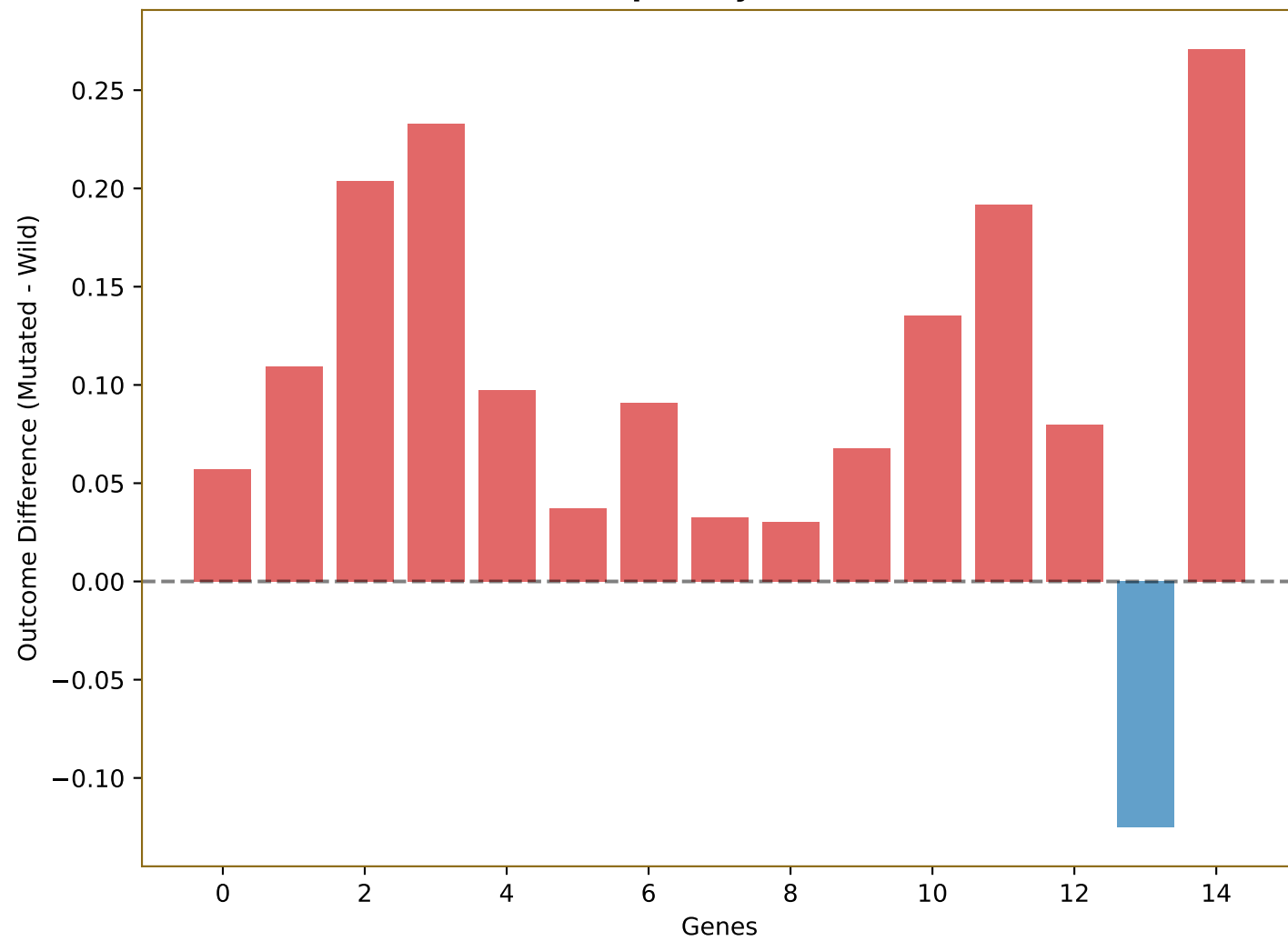
Radiomics-Gene Correlation (Gene_01)



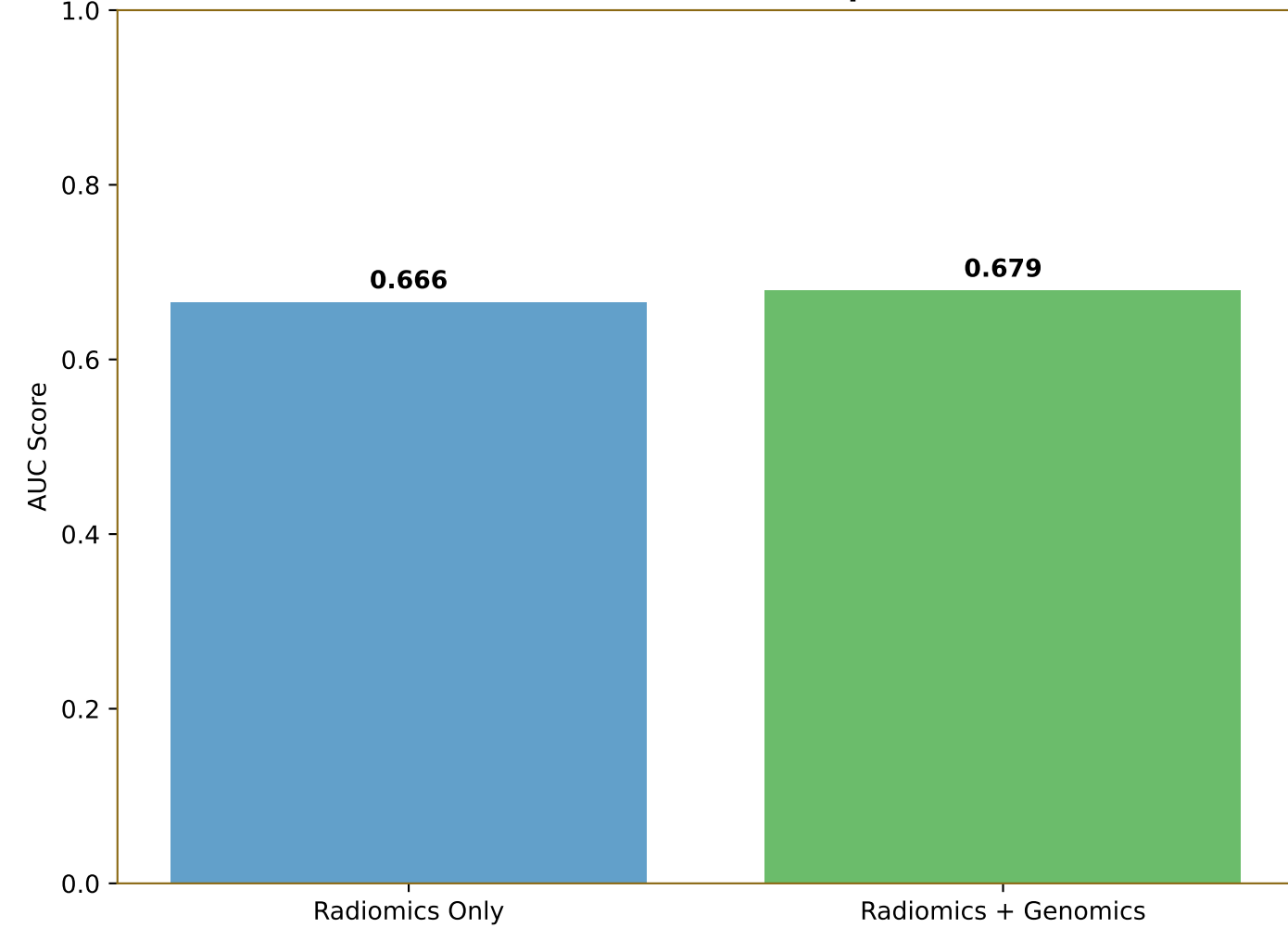
Gene Mutation Frequency



Outcome Impact by Gene Mutation

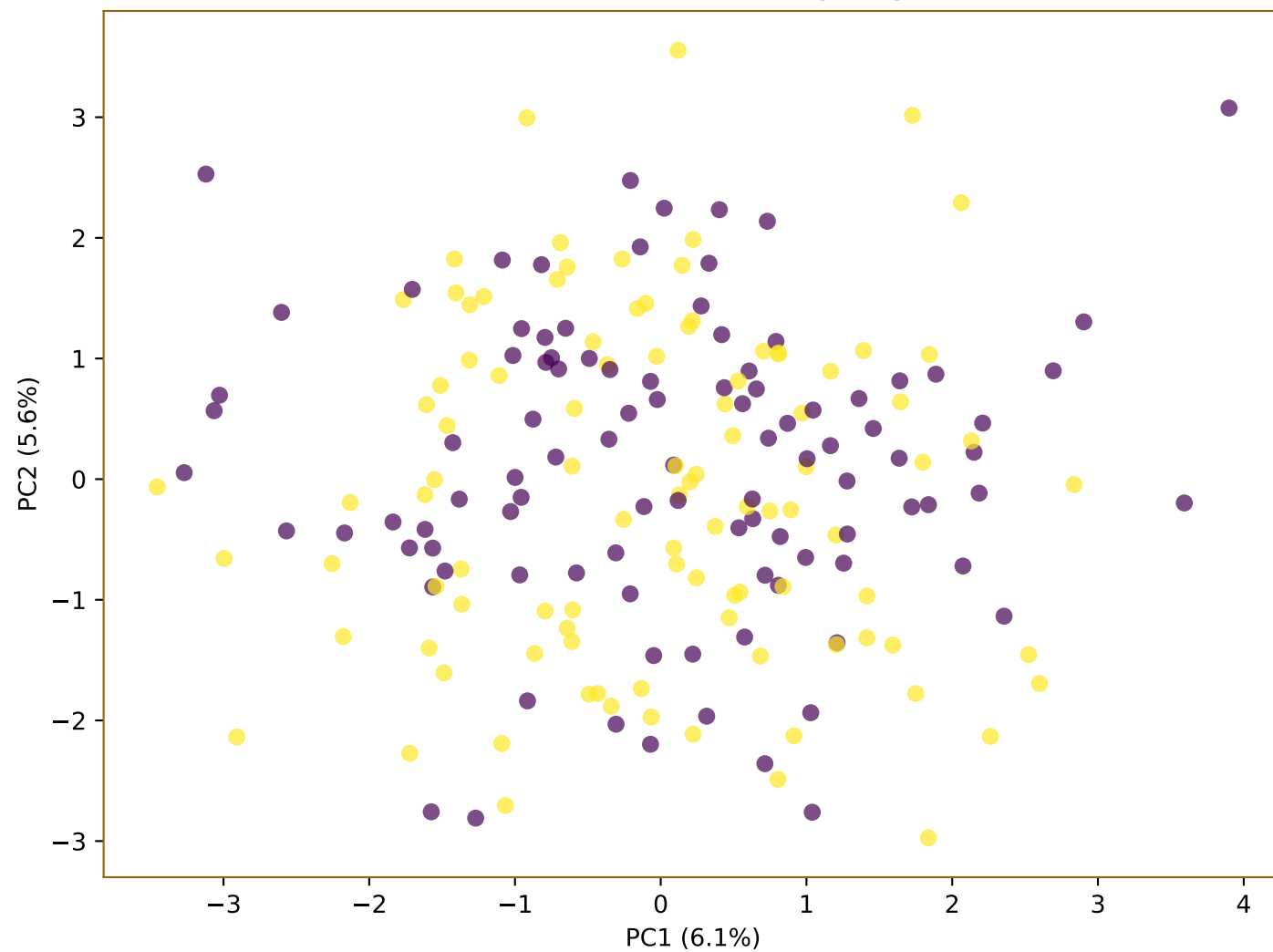


Model Performance Comparison

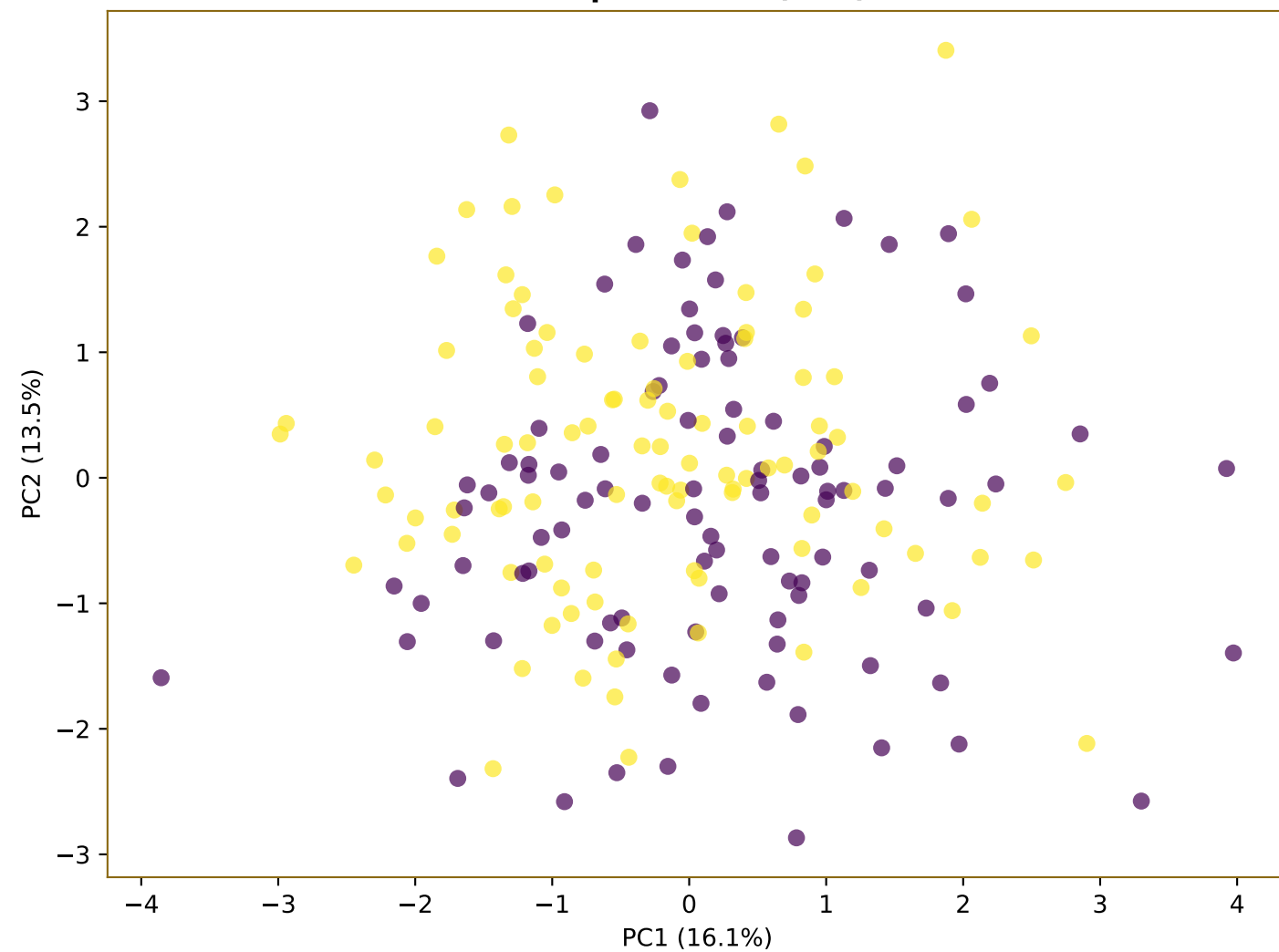


Deep Radiomics & Feature Learning

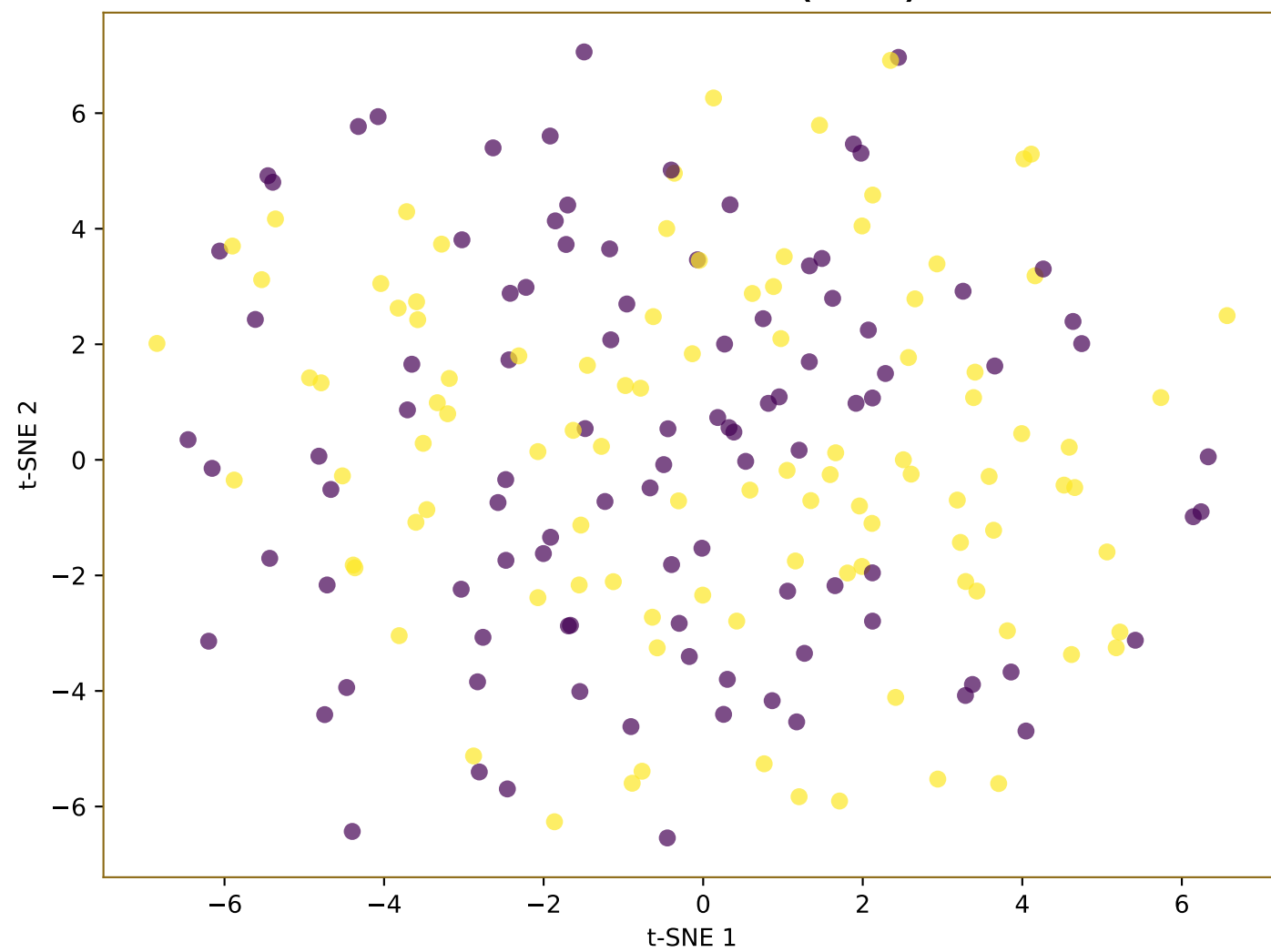
Radiomics Features (PCA)



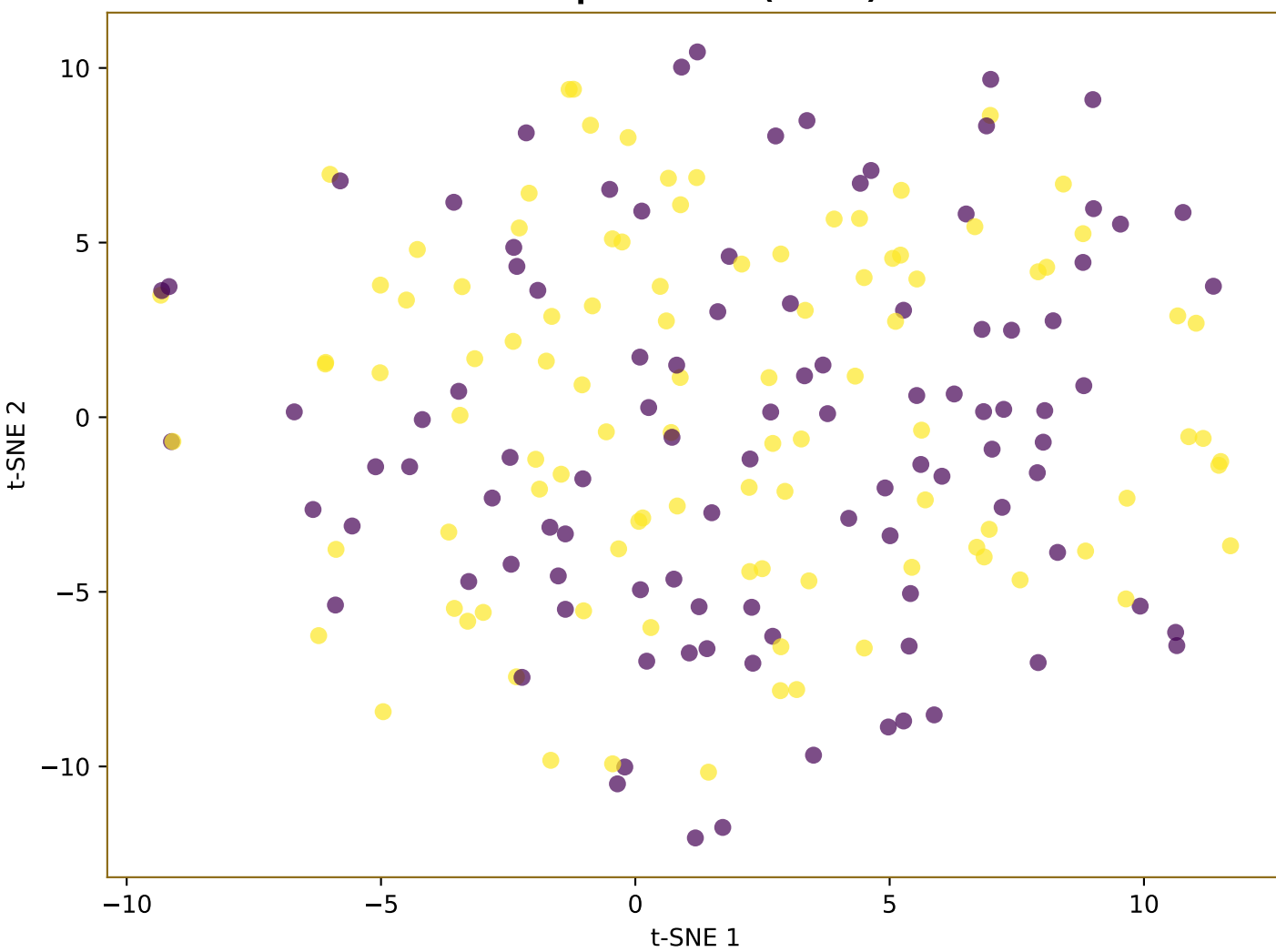
Deep Features (PCA)



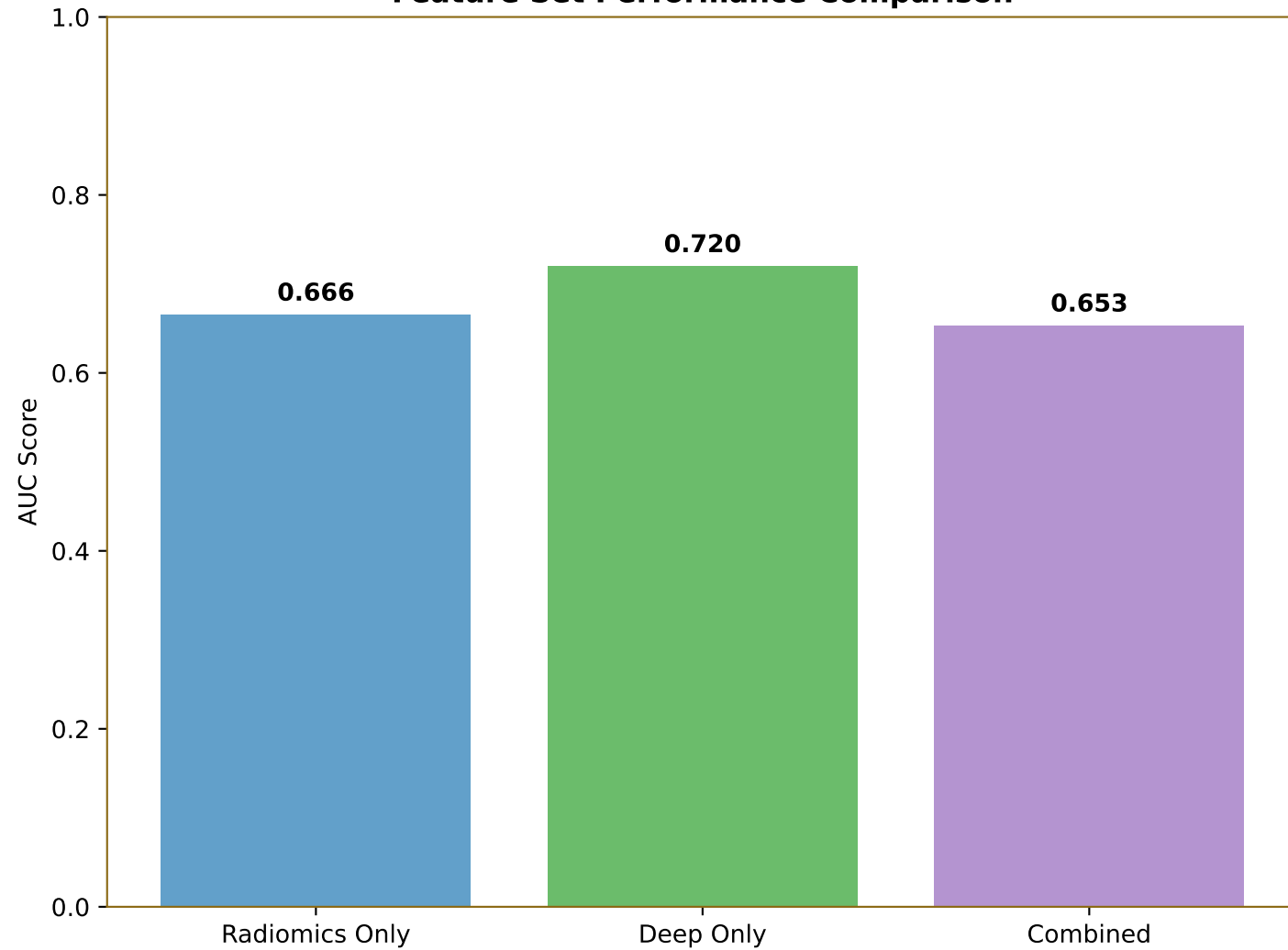
Radiomics Features (t-SNE)



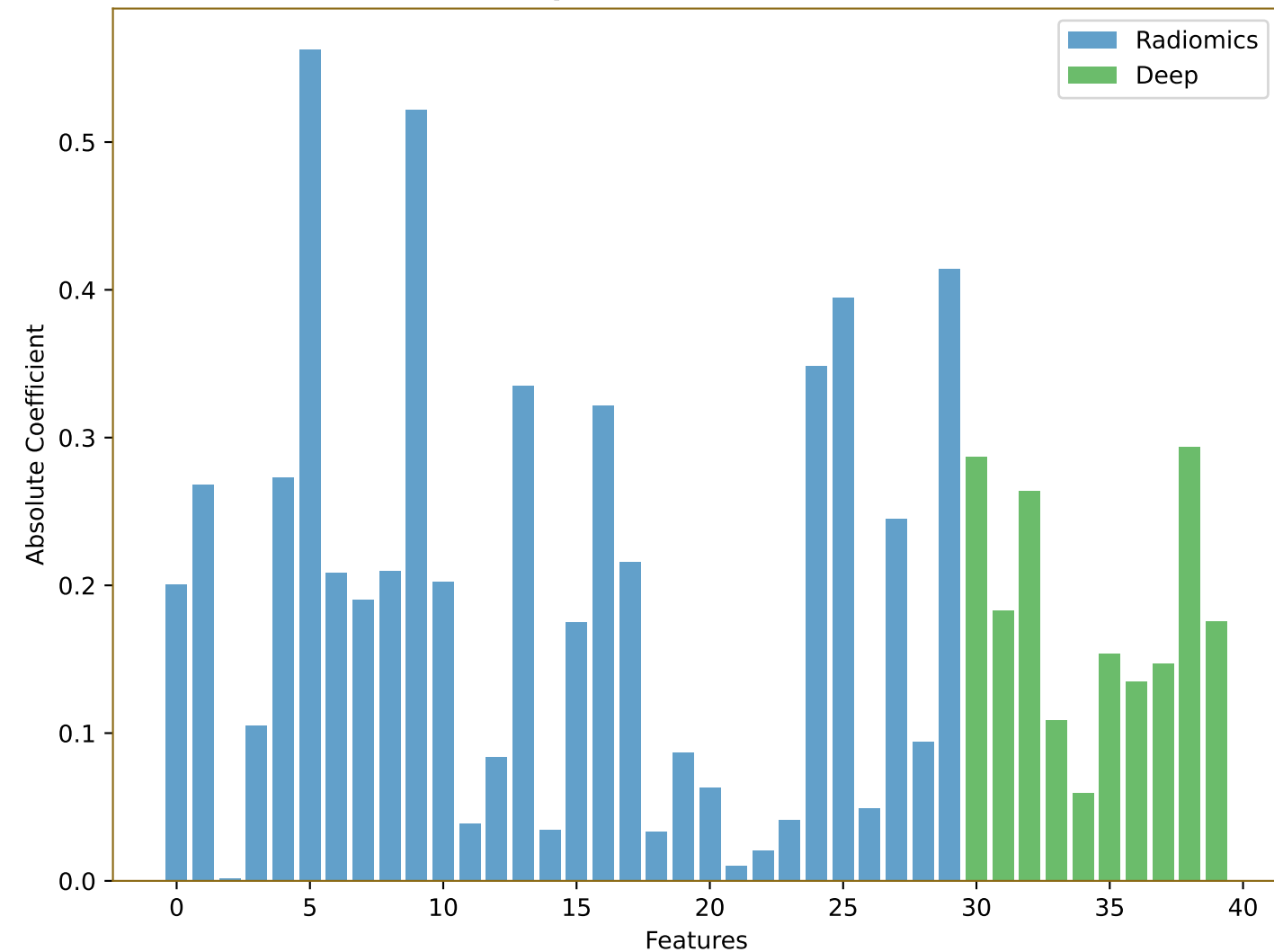
Deep Features (t-SNE)



Feature Set Performance Comparison

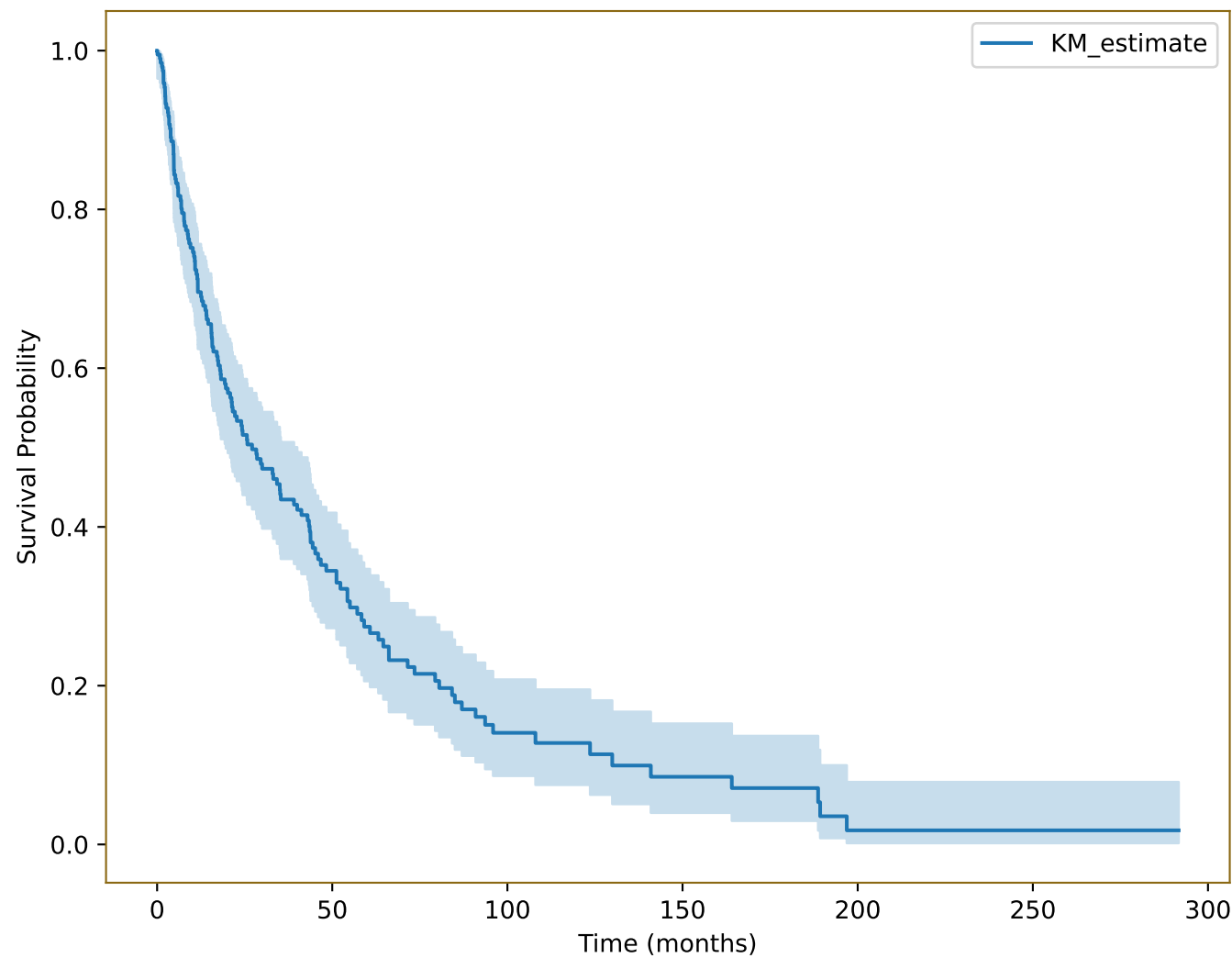


Feature Importance in Combined Model

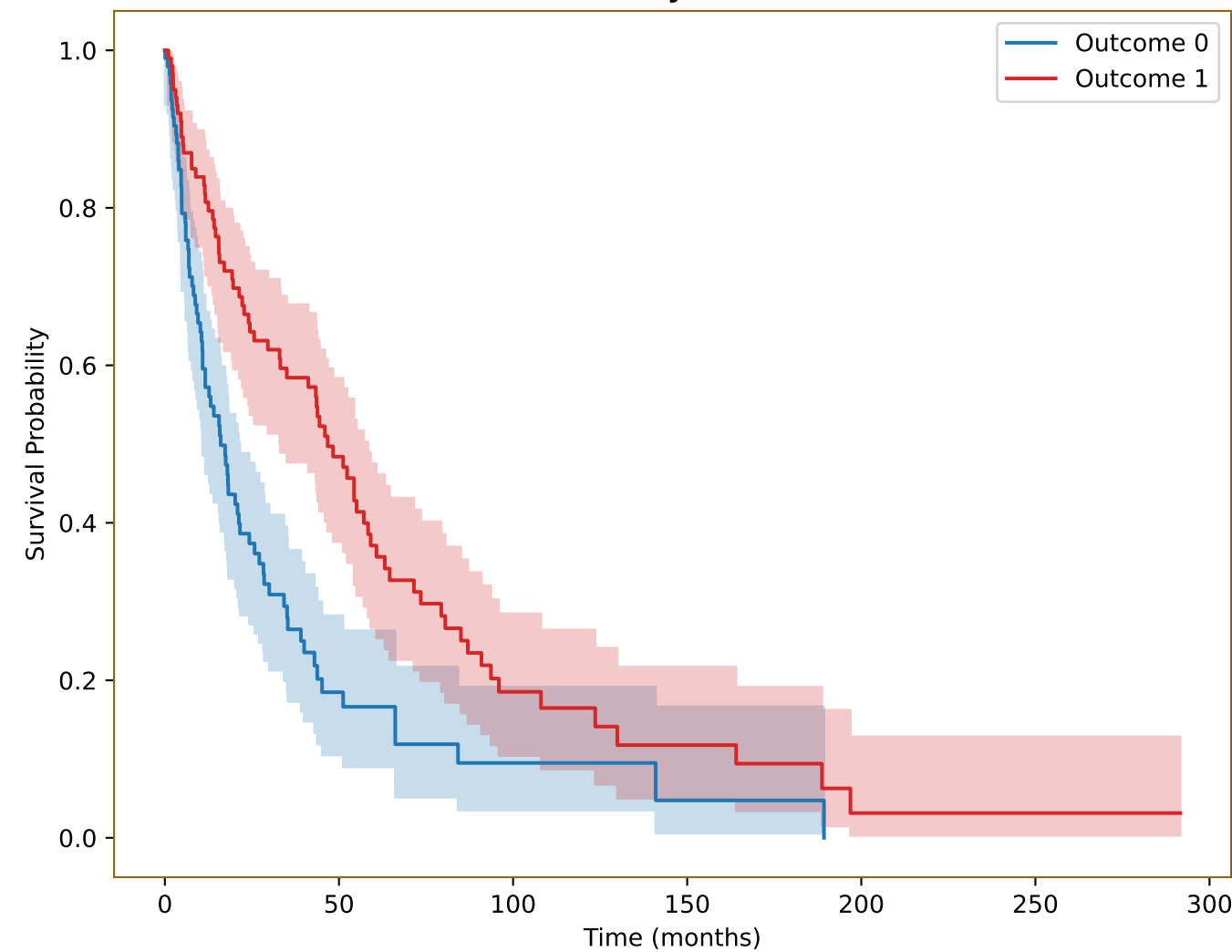


Survival & Time-to-Event Analysis

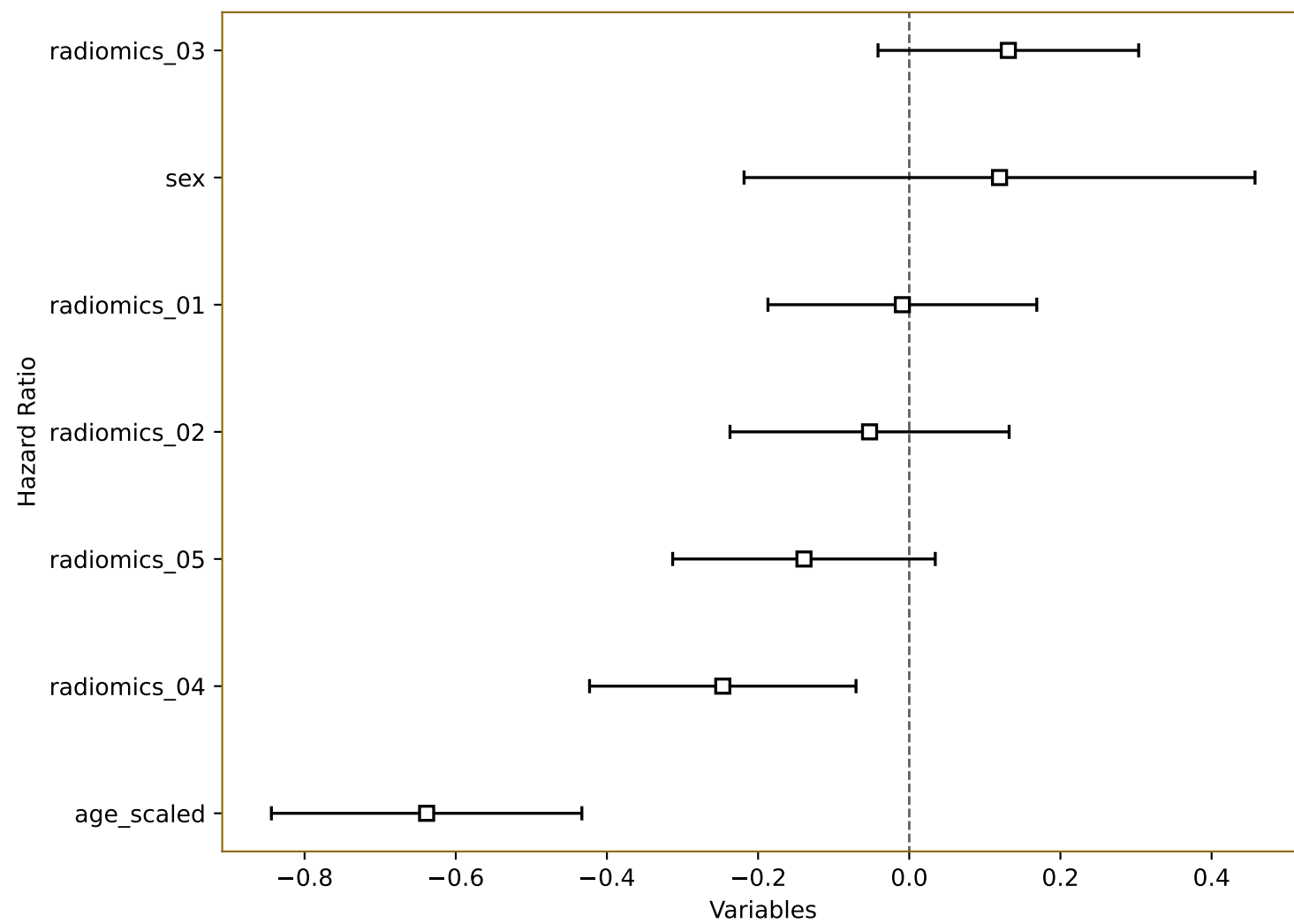
Overall Survival



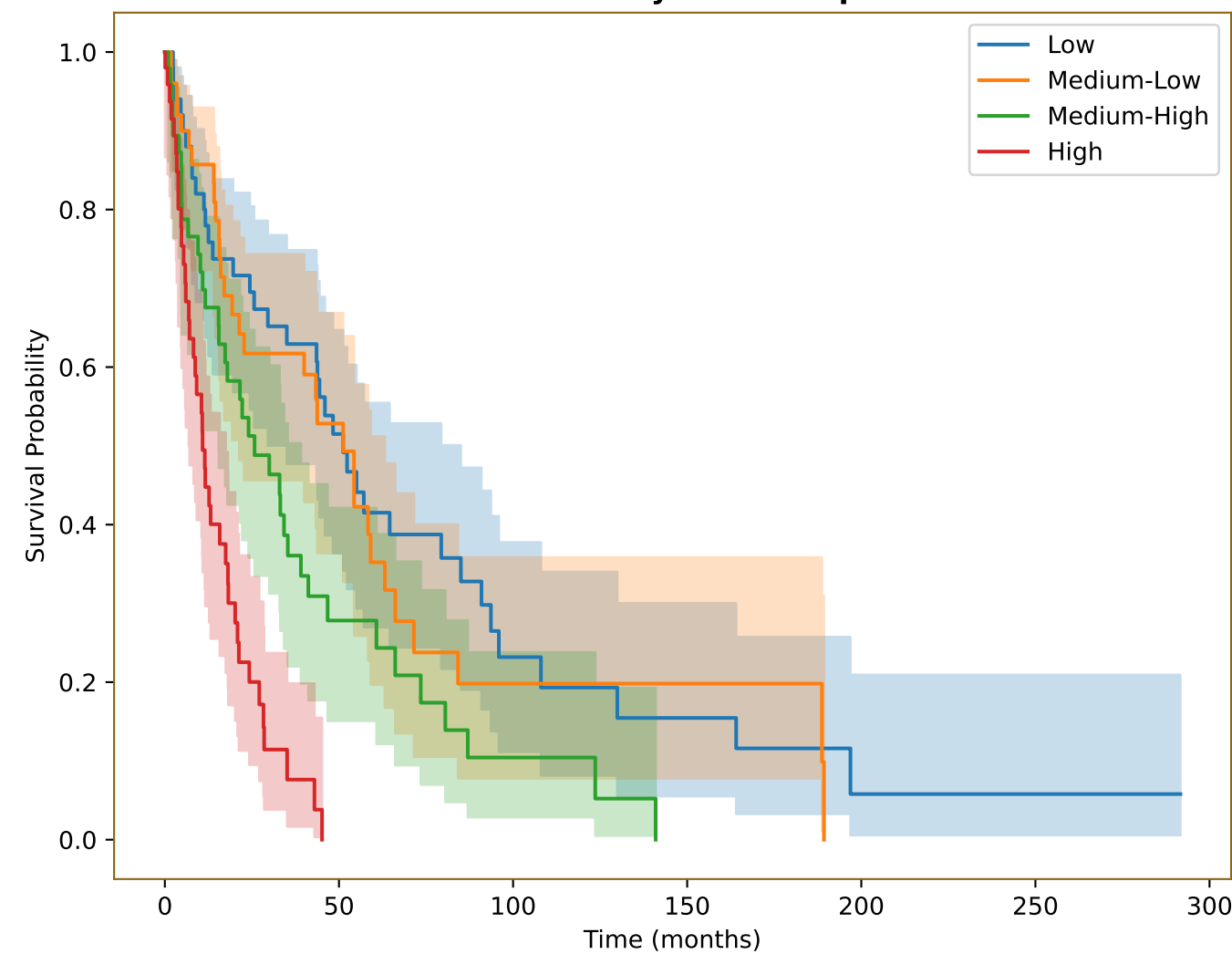
Survival by Outcome



Cox Model: Hazard Ratios

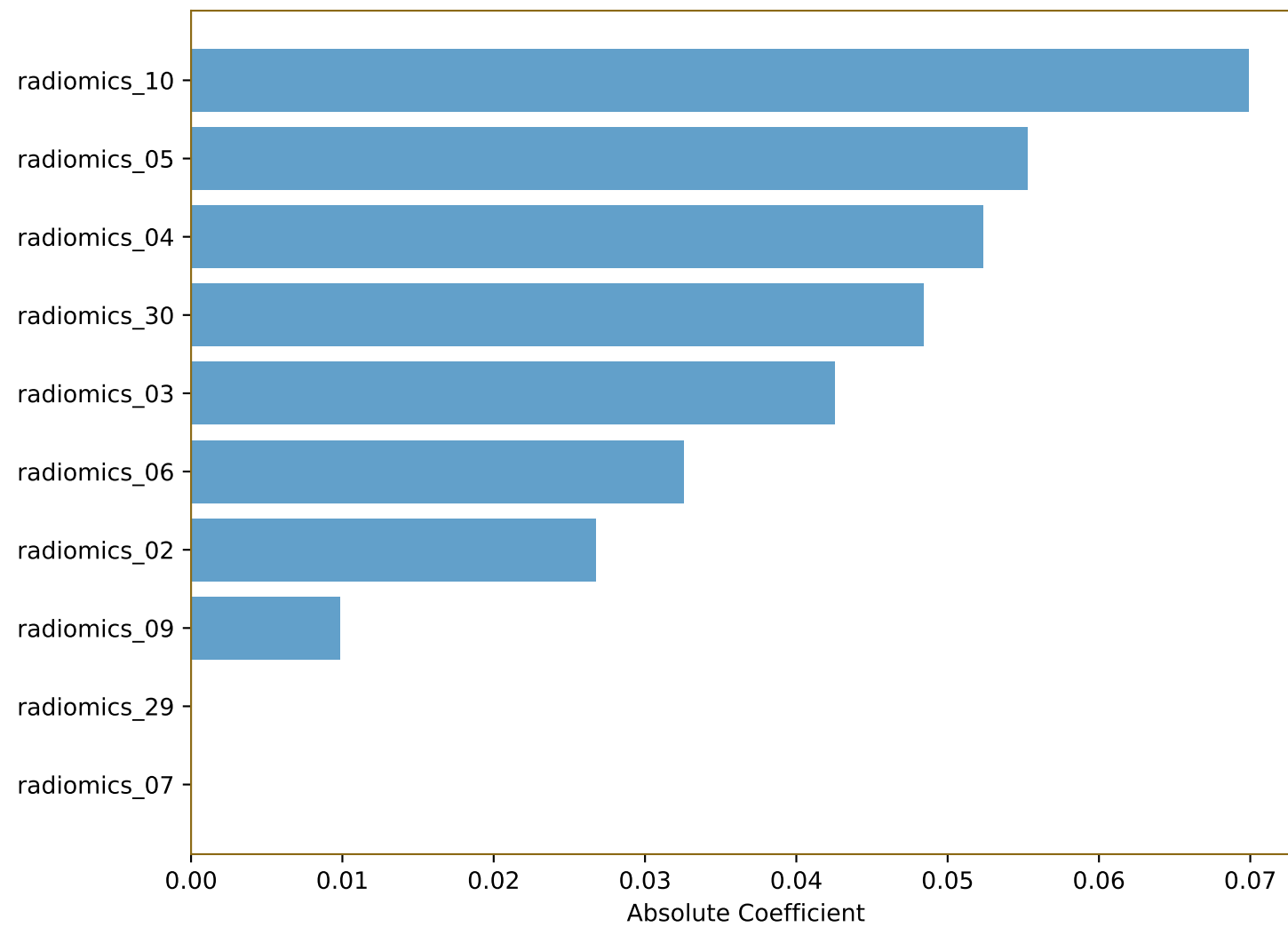


Survival by Risk Group

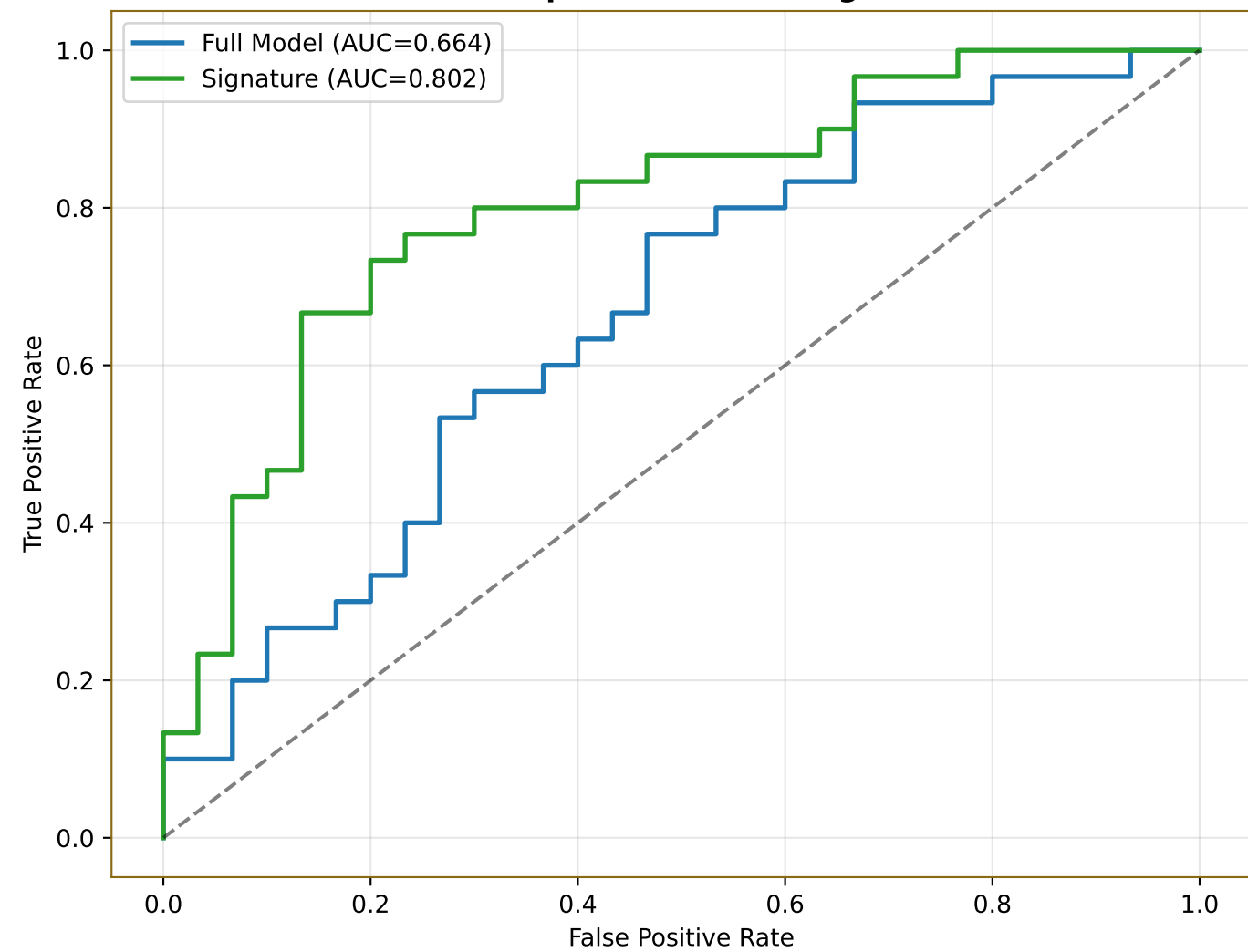


Radiomics Signature Development

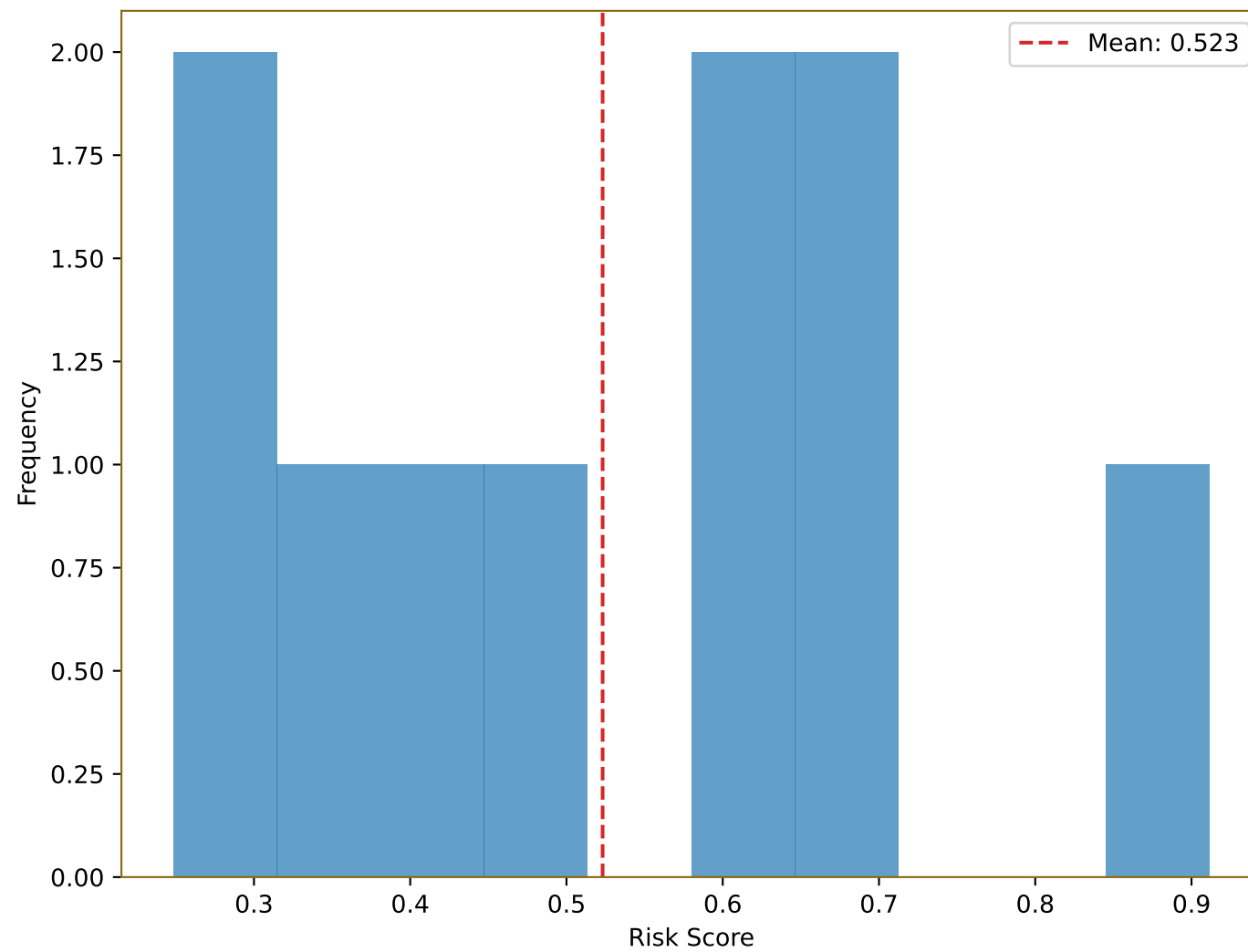
LASSO Feature Selection



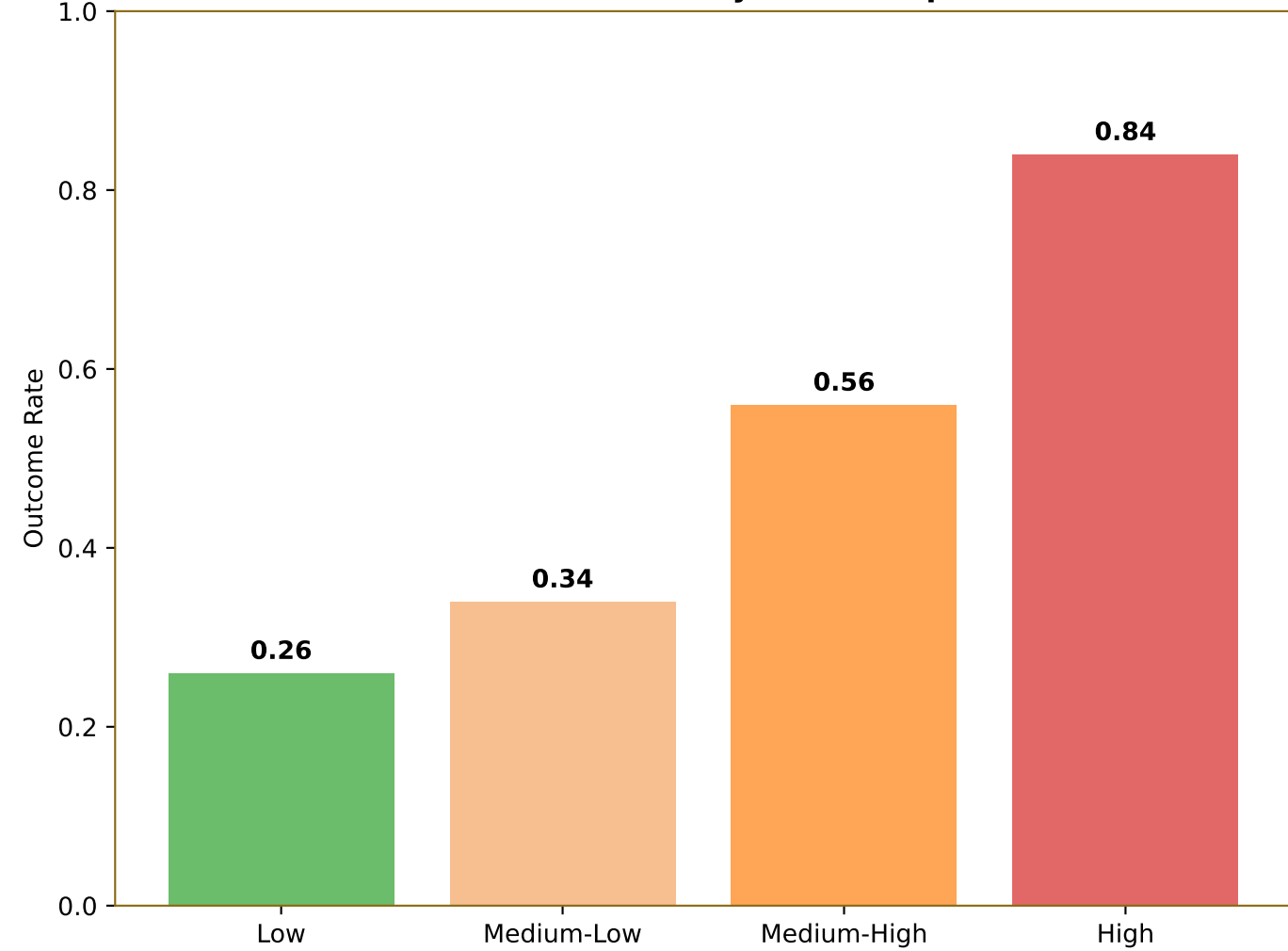
ROC Comparison: Full vs Signature



Risk Score Distribution

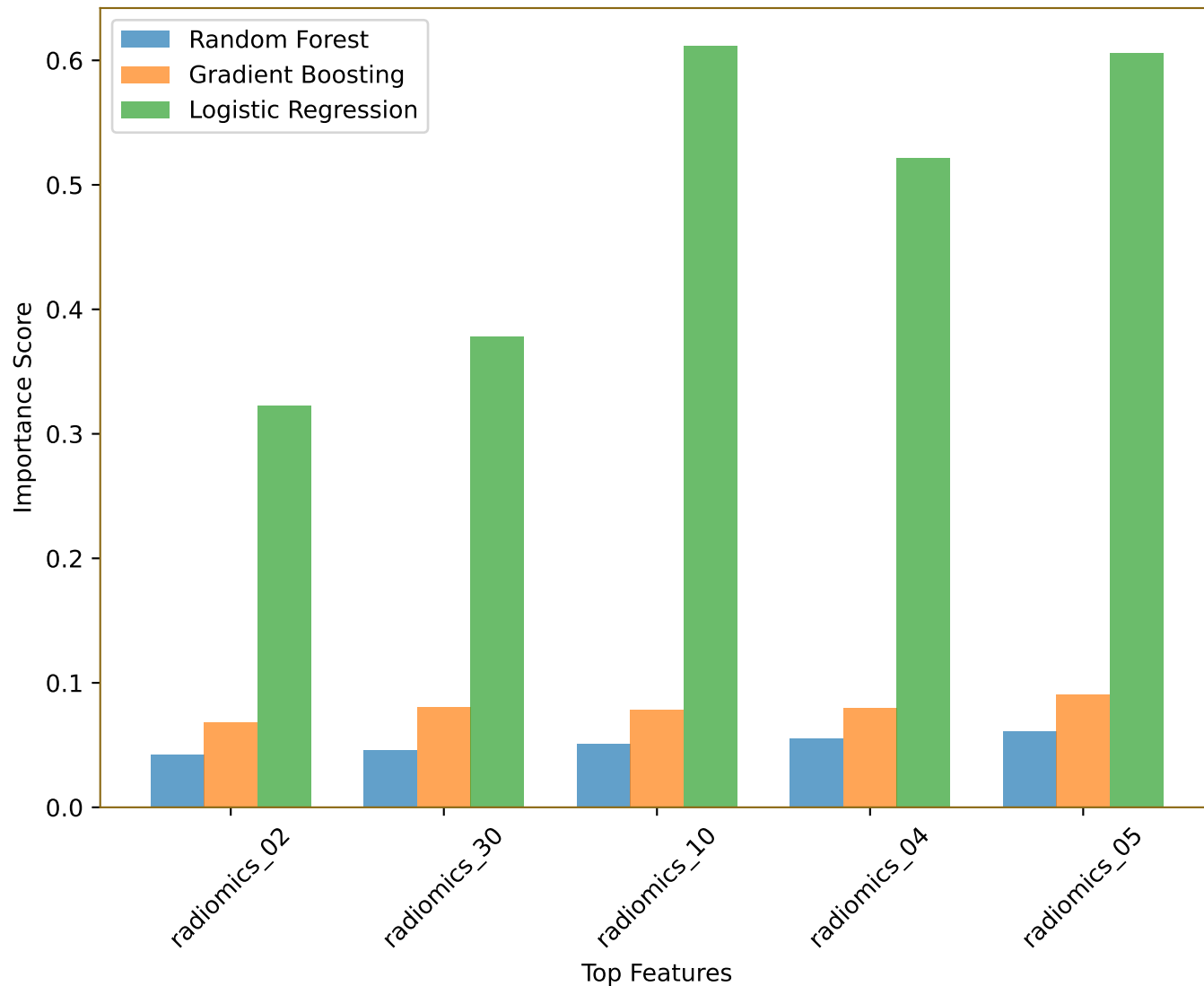


Outcome Rate by Risk Group

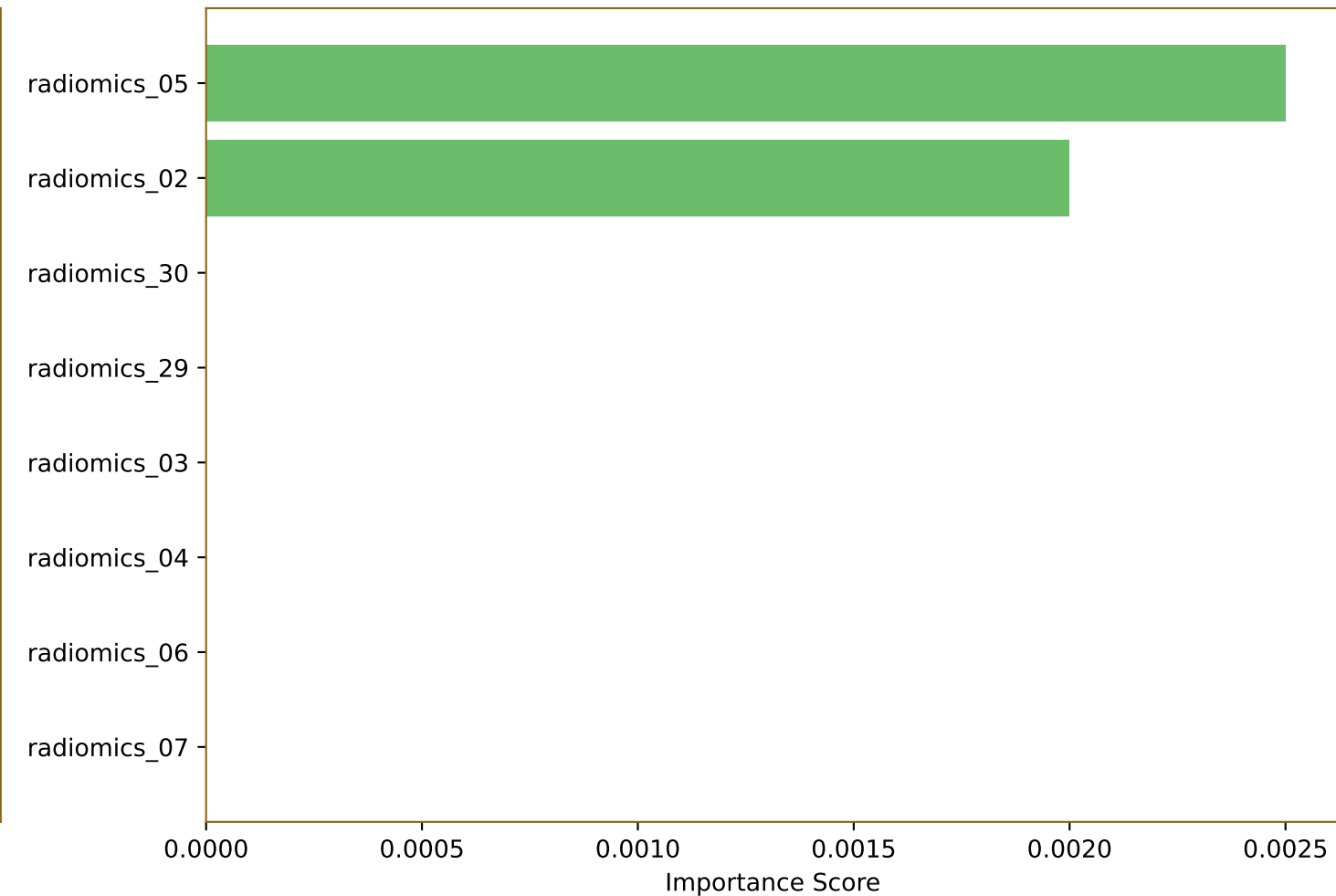


Explainable AI & Model Interpretability

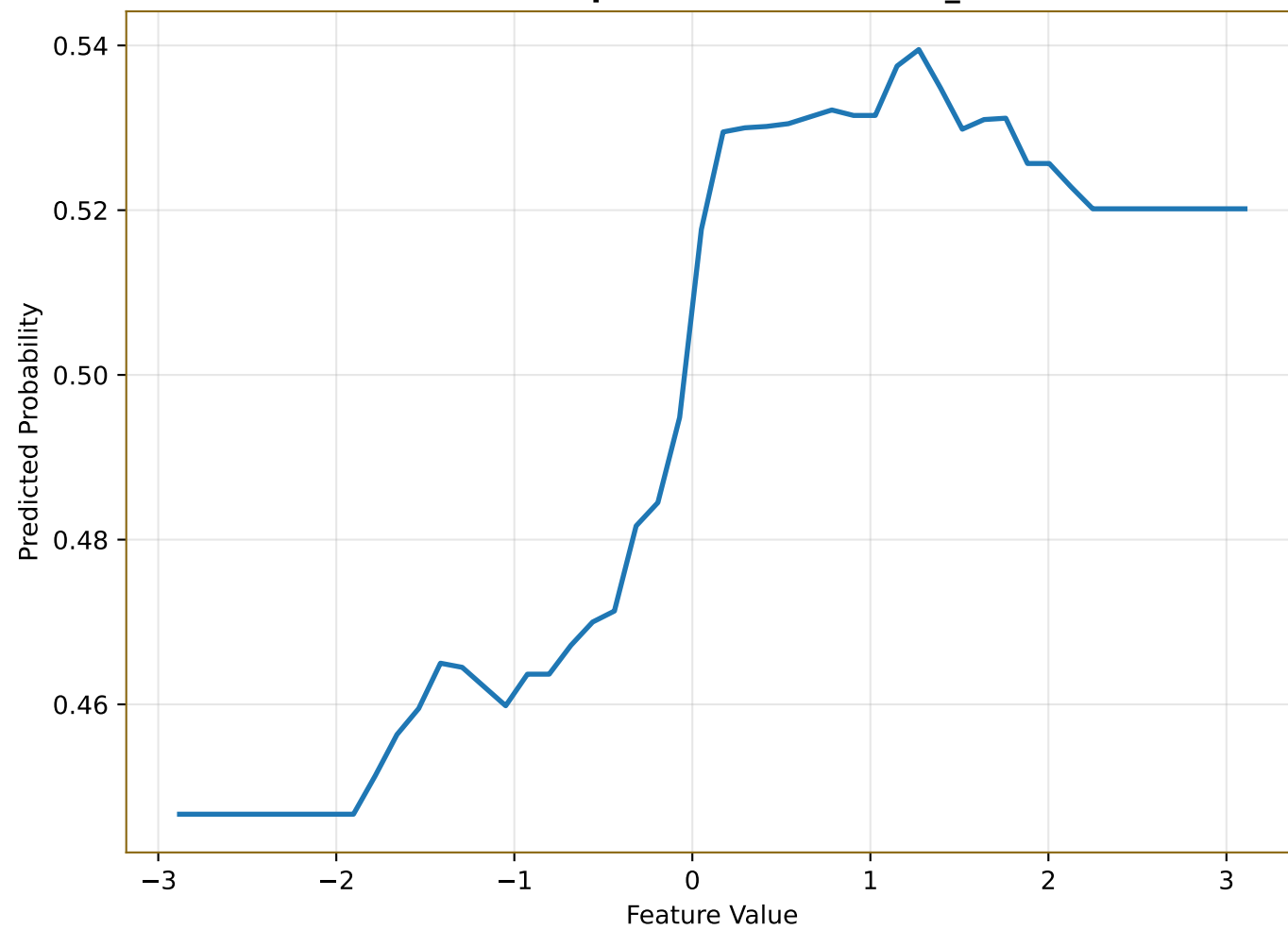
Feature Importance Comparison



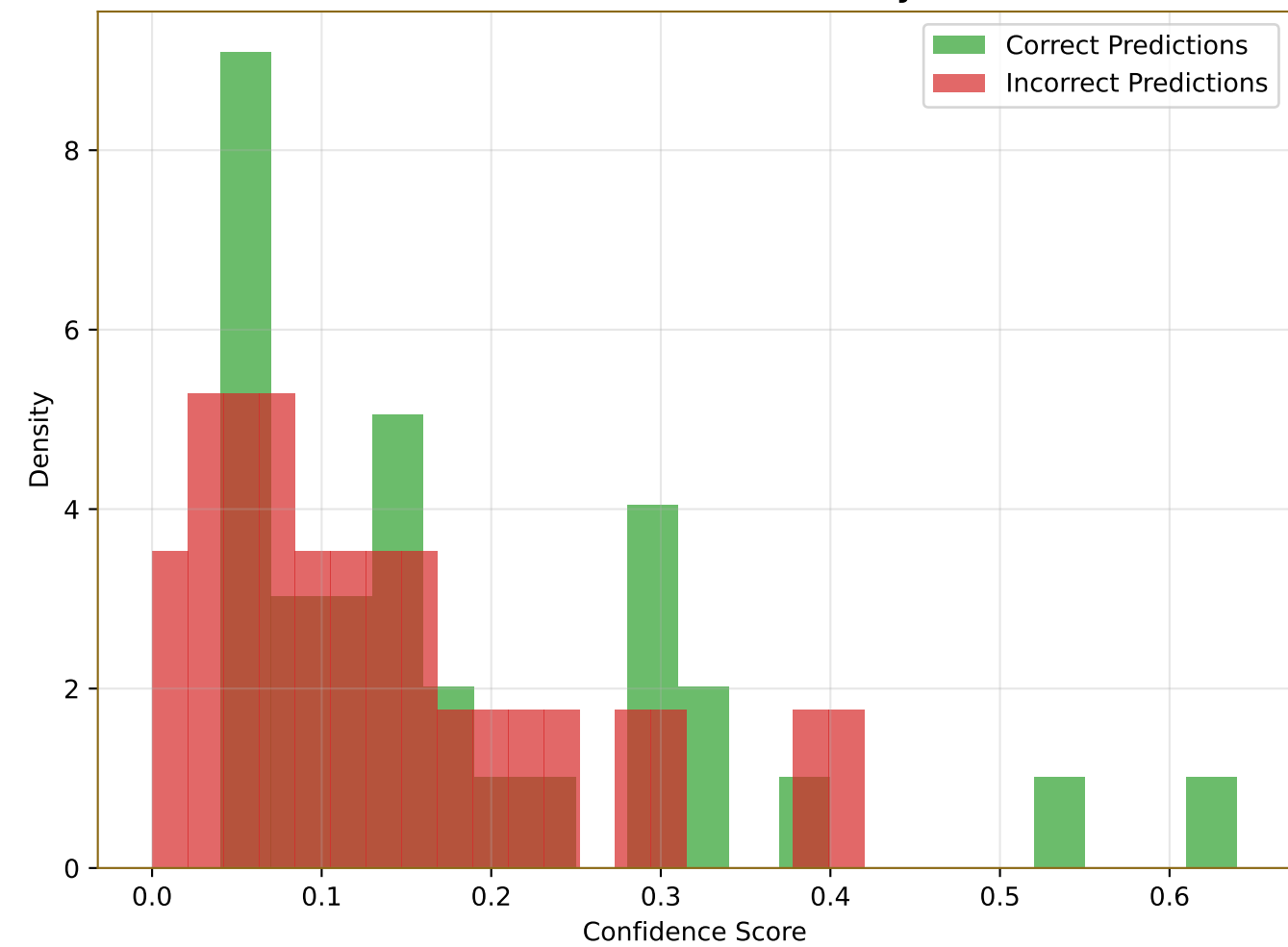
Permutation Importance



Partial Dependence: radiomics_05

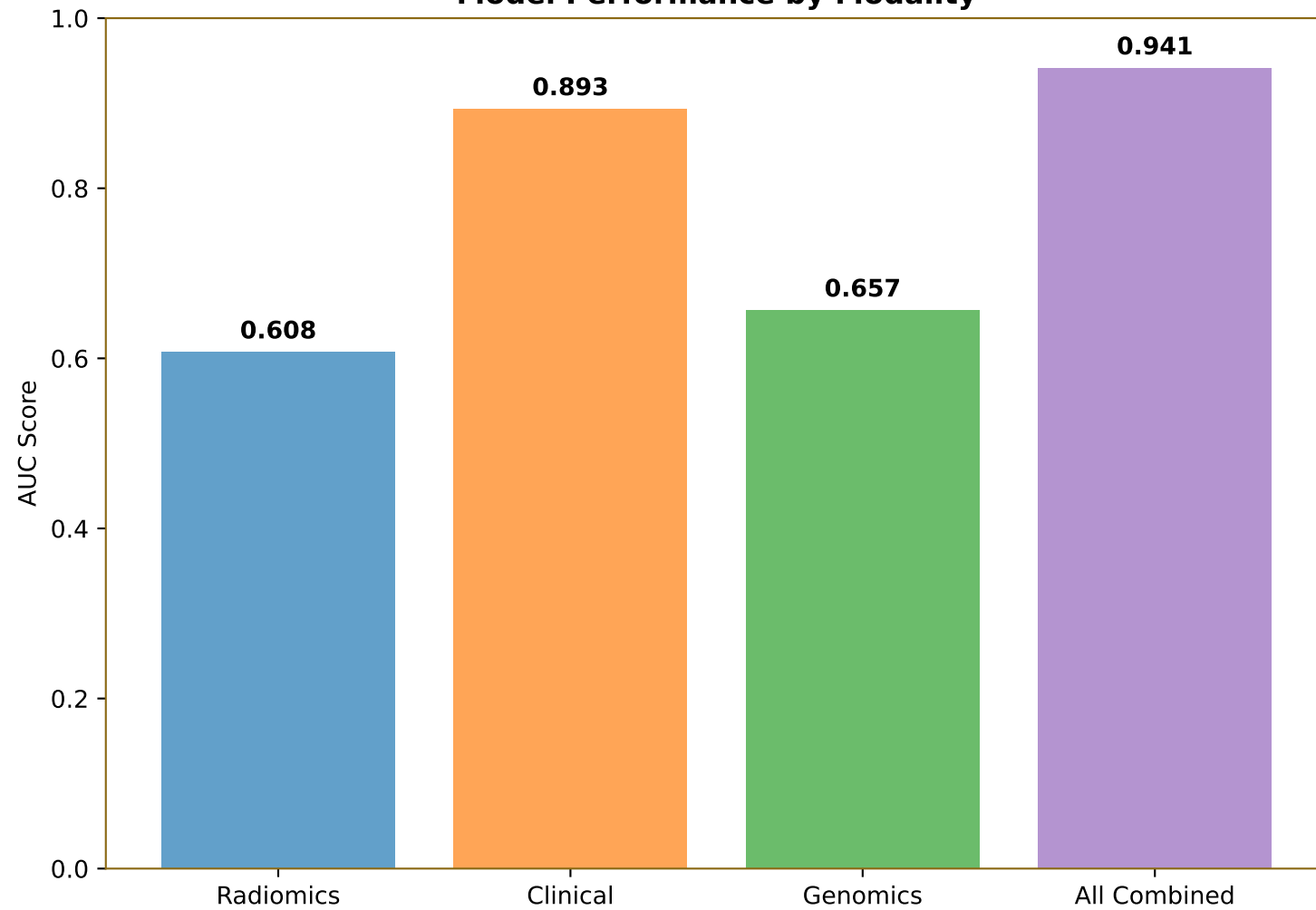


Prediction Confidence Analysis

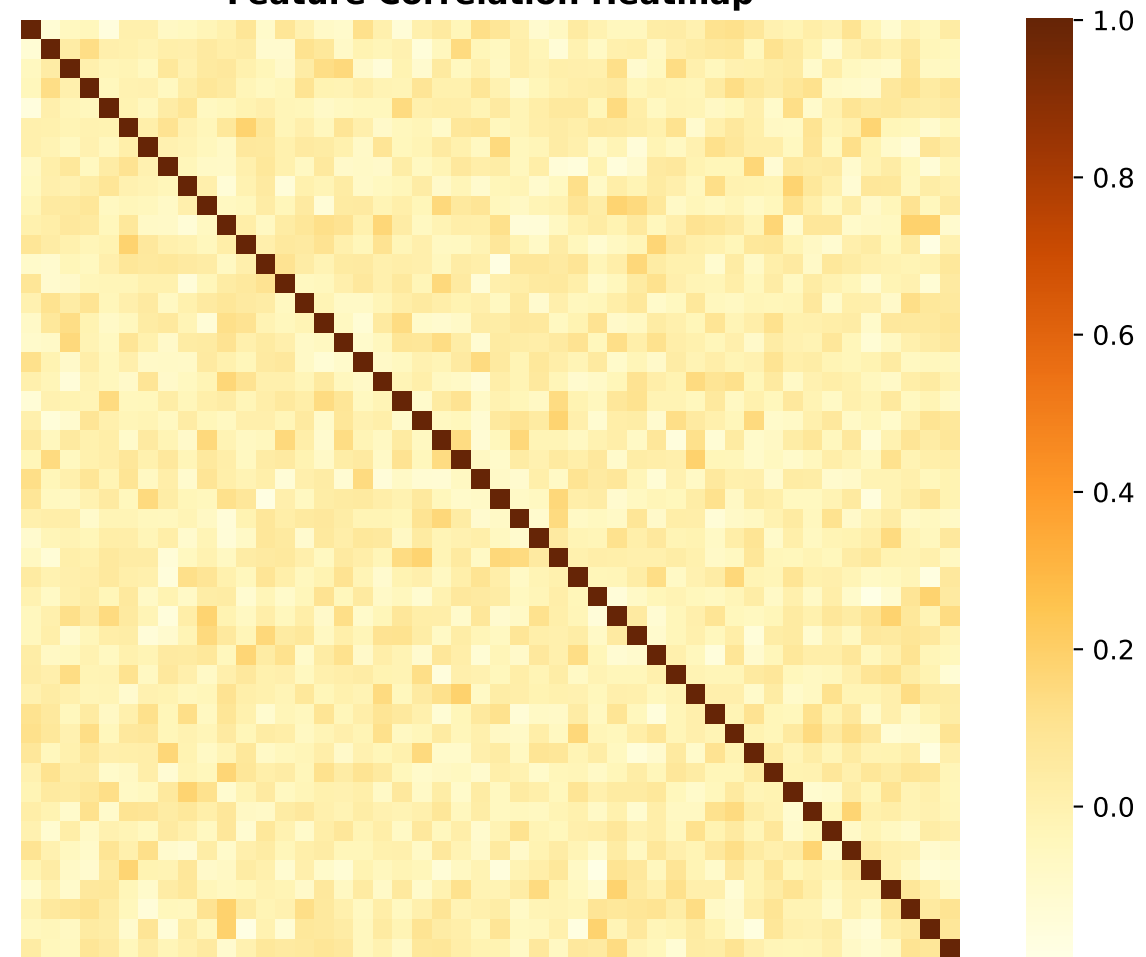


Multi-Modal Data Integration

Model Performance by Modality

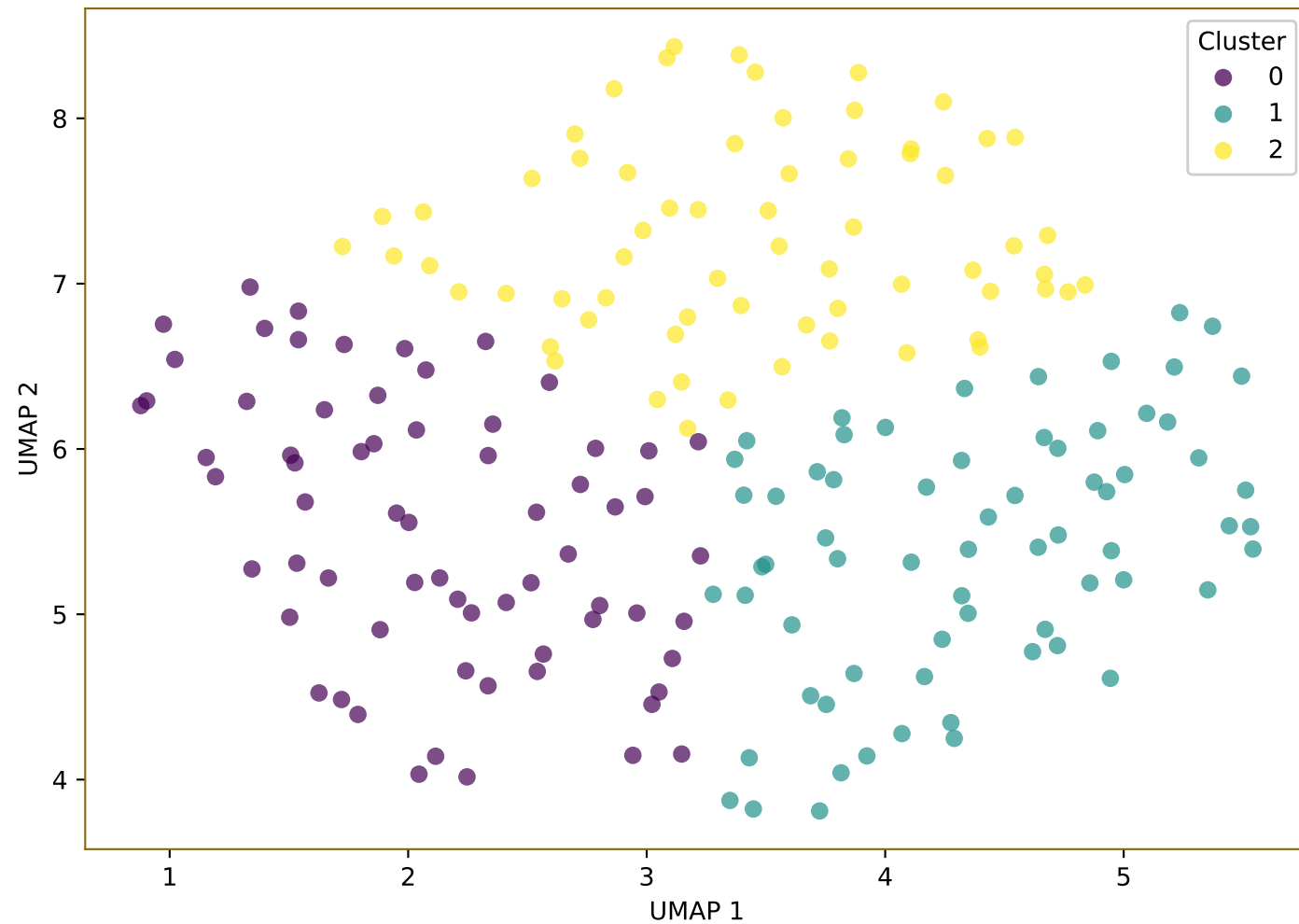


Feature Correlation Heatmap

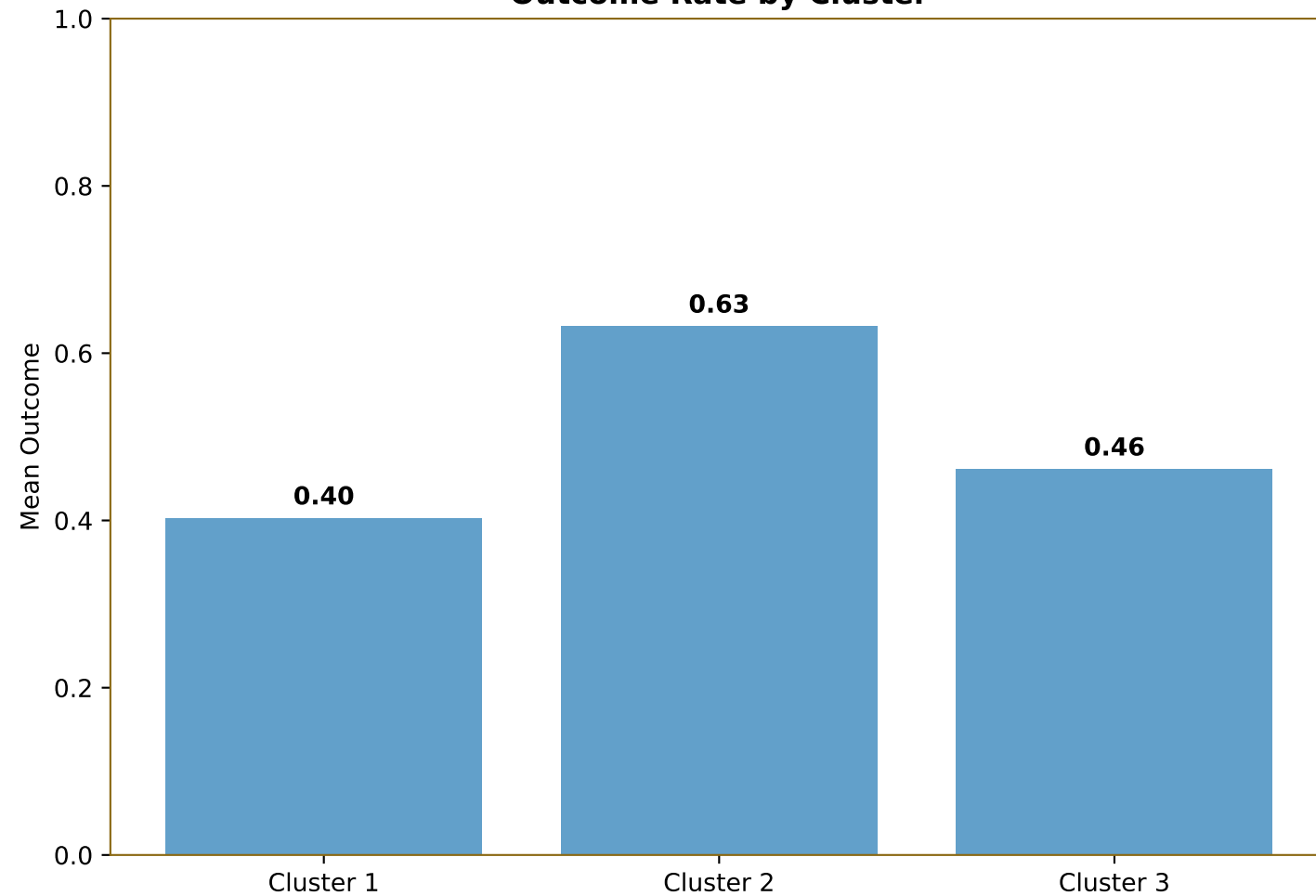


Clustering & Phenotyping

UMAP Embedding with K-means Clusters



Outcome Rate by Cluster



[Longitudinal & Delta Radiomics]
Feature change over time, delta analysis
(placeholder - to be implemented)

[Feature Robustness & Harmonization]
ComBat, multi-center harmonization
(placeholder - to be implemented)

[Clinical Trial Integration]
Stratification, enrichment, trial design
(placeholder - to be implemented)